

Taller

Introducción a la detección de objetos espaciales
con Python e inteligencia artificial

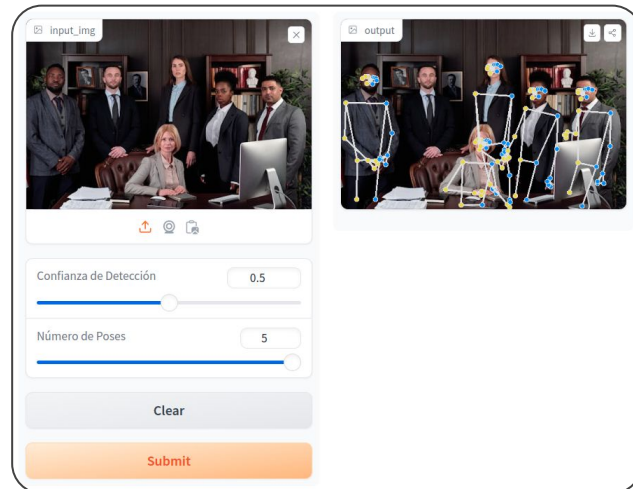
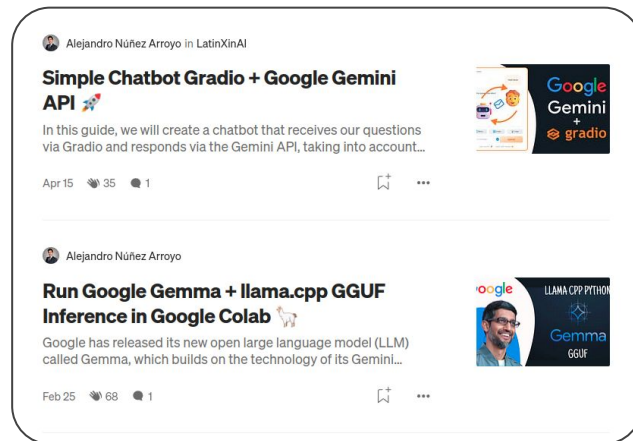
Ing. Alejandro Núñez Arroyo

¡Hola Mundo!

- Ingeniero Mecatrónico
- Investigador de AI en Fogsphere
- Organizador en el Google Developers Group Sucre

@alenunezarroyo

beacons.ai/aleneunezarroyo



Cronograma de actividades

Clase 1: 26 de abril

Actividad teórica:

- Introducción a Computer Vision
- Tareas de Computer Vision
- Flujo de trabajo en un proyecto de Computer Vision
- Datasets para la detección de objetos espaciales
- Preparación de datos

Actividad práctica:

- Introducción a Google Colab y Python
- Componentes de una imagen

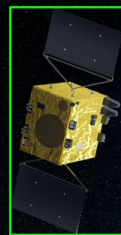
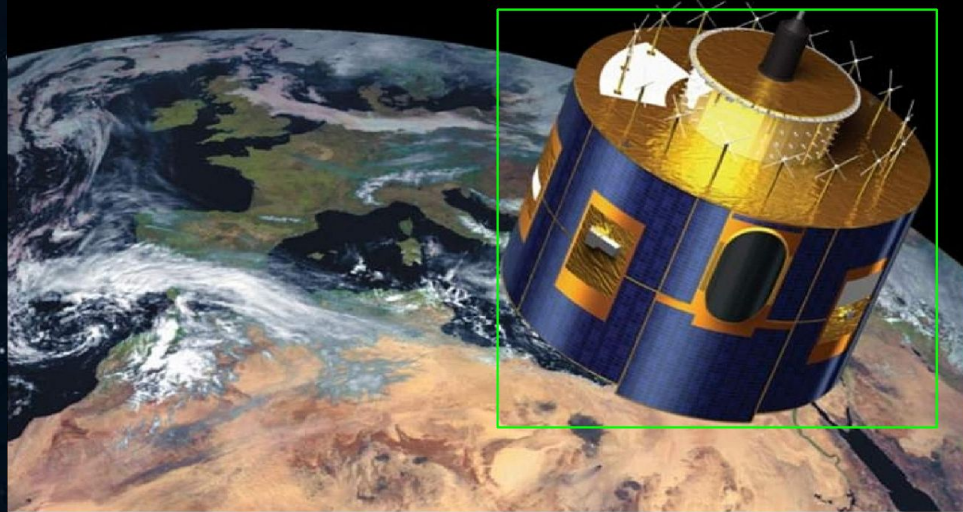
Clase 2: 3 de mayo

Actividad teórica:

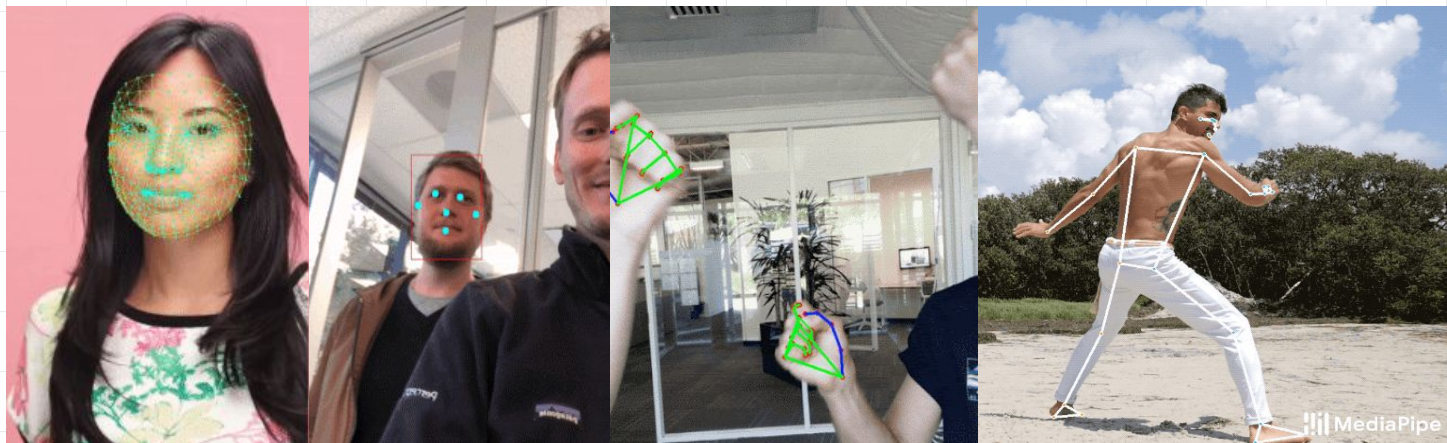
- ¿Cómo funciona la detección de objetos?
- Modelos de clasificación de objetos (RF-DETR, YOLO, RetinaNet, etc)

Actividad práctica:

- Preparación de datos
- Entrenamiento de modelos de clasificación de objetos espaciales
- Evaluación y consumo del modelo entrenado
- Conclusiones y trabajo a futuro



¿Qué tan inteligentes son los modelos de Computer Vision?



Ejemplo de clasificación (Entrenamiento)



0



1



1

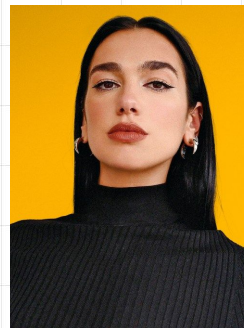


0

Ejemplo de clasificación (Su turno)



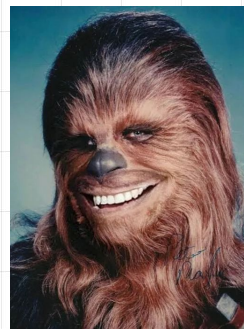
a)



b)



c)



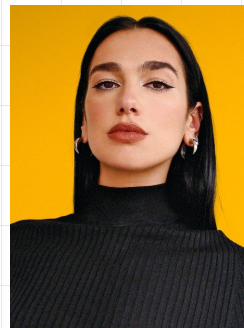
d)

Ejemplo de clasificación (Modelo entrenado)



1

a)



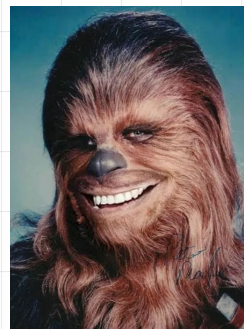
0

b)



1

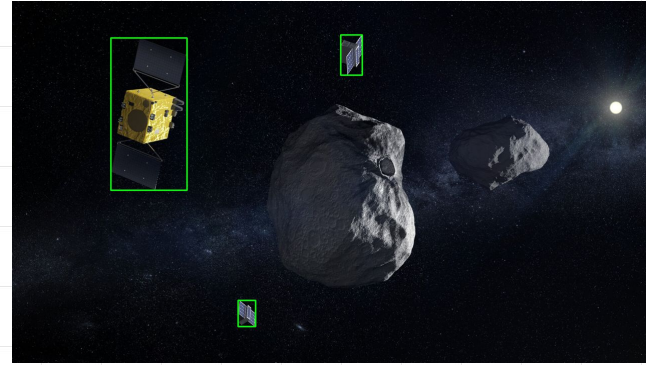
c)



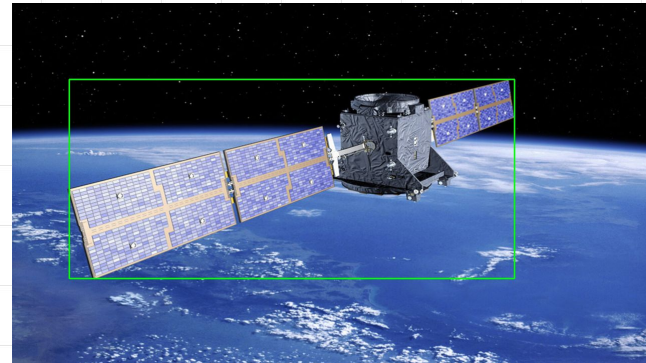
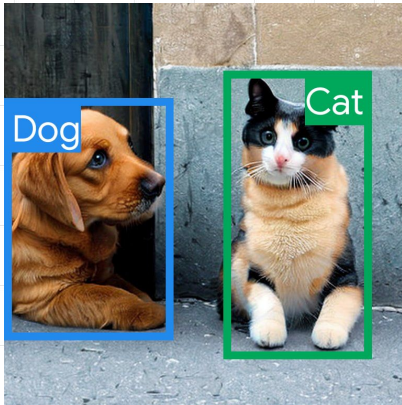
1

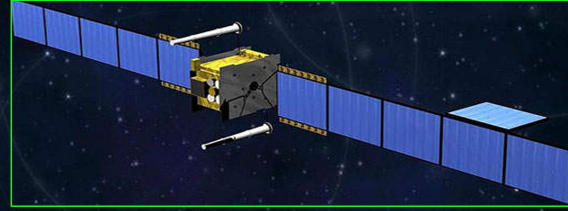
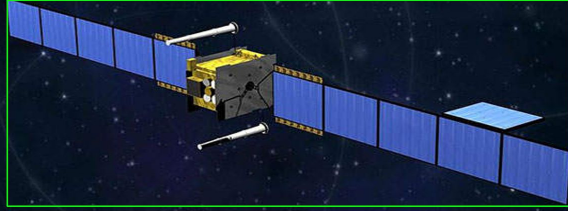
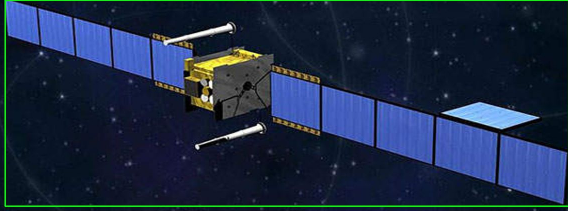
d)

¿Por qué se entrena?

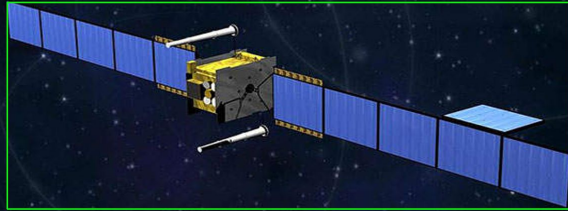
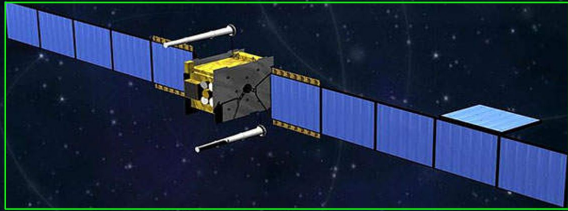


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¿Es fácil crear un modelo de Computer Vision?

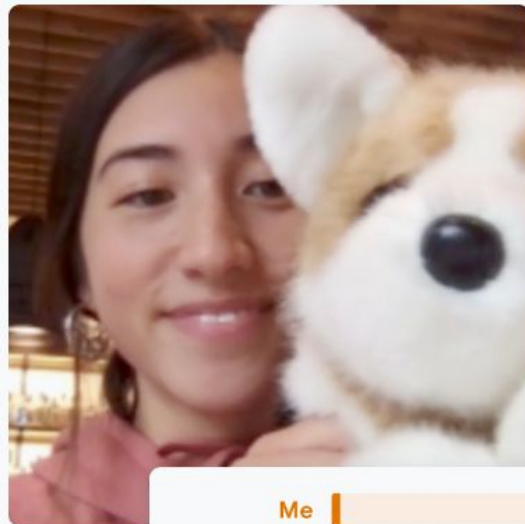


Teachable Machine

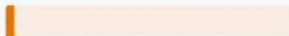
Train a computer to recognize your own images, sounds, & poses.

A fast, easy way to create machine learning models for your sites, apps, and more – no expertise or coding required.

Get Started



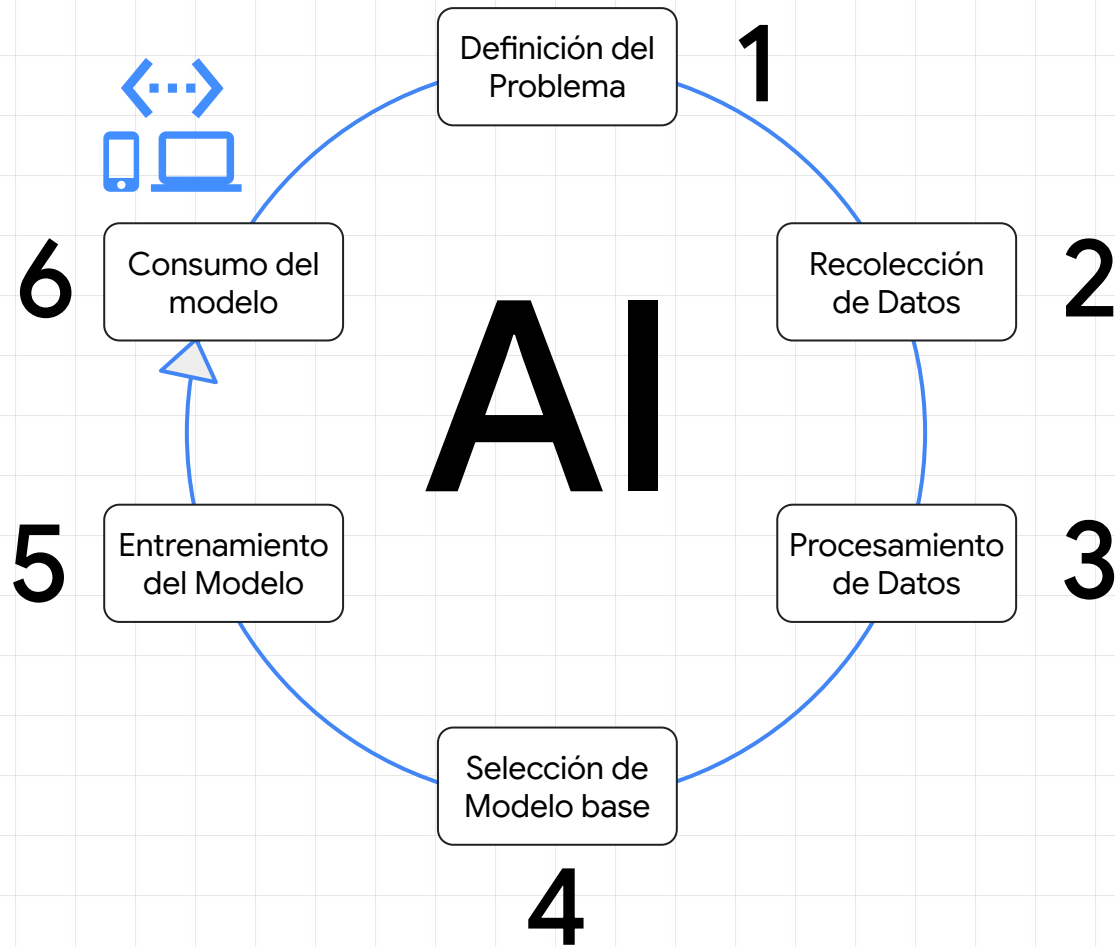
Me

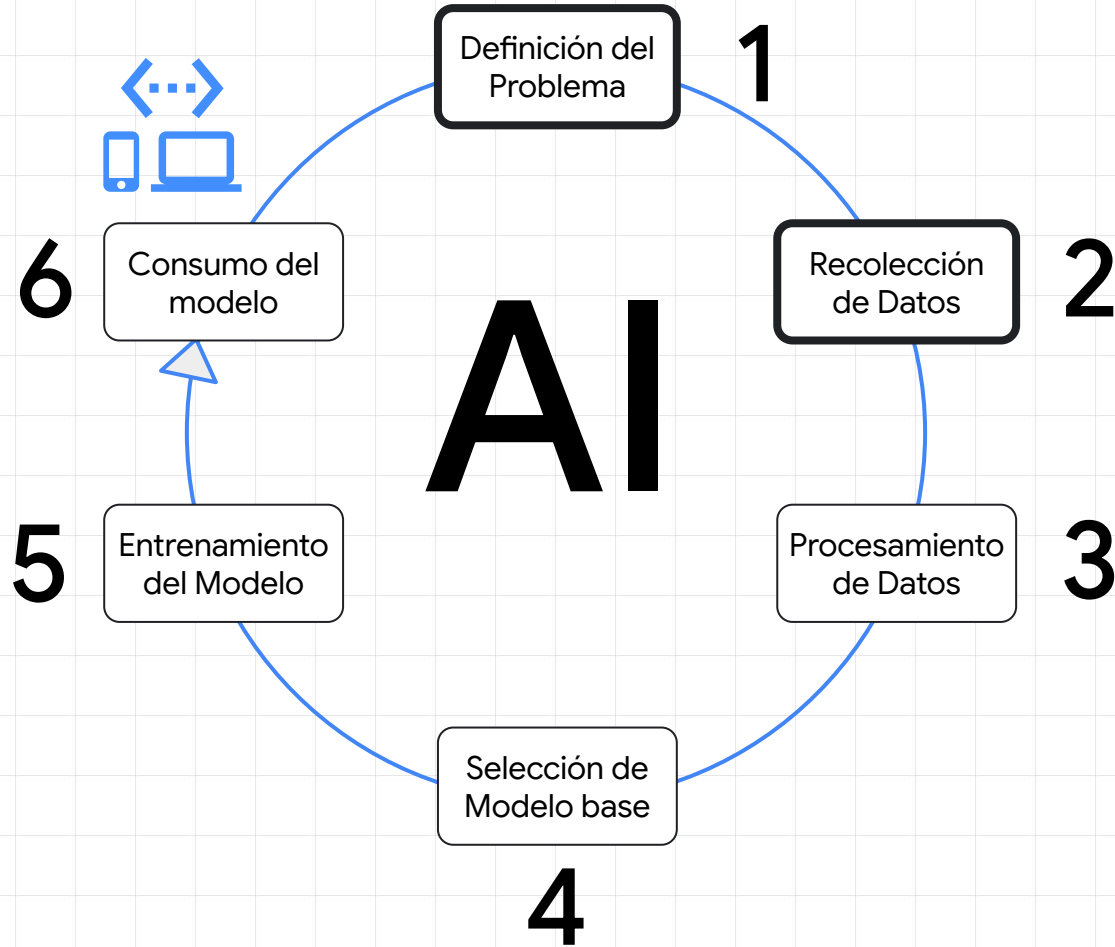


Me + Dog <3

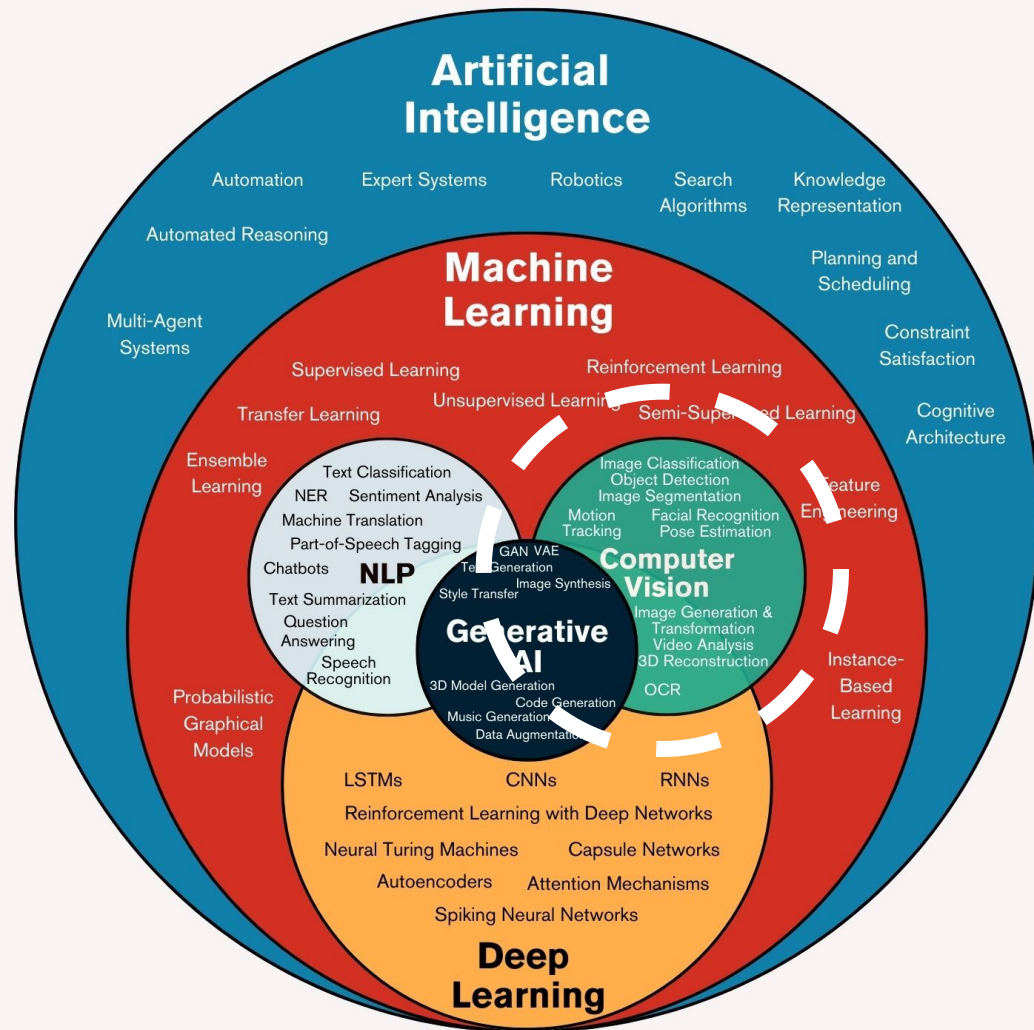


98%



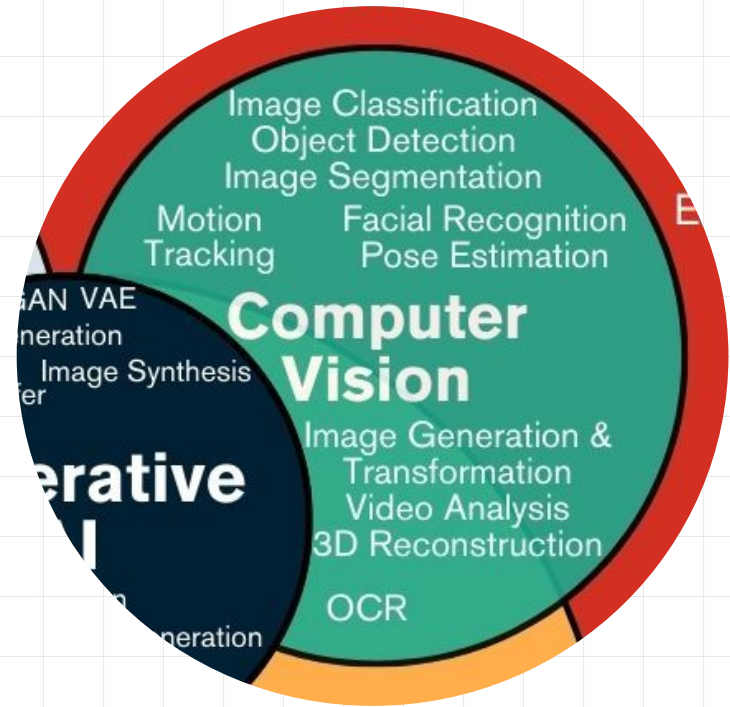


¿Computer Vision es Inteligencia Artificial?



Tareas en Computer Vision

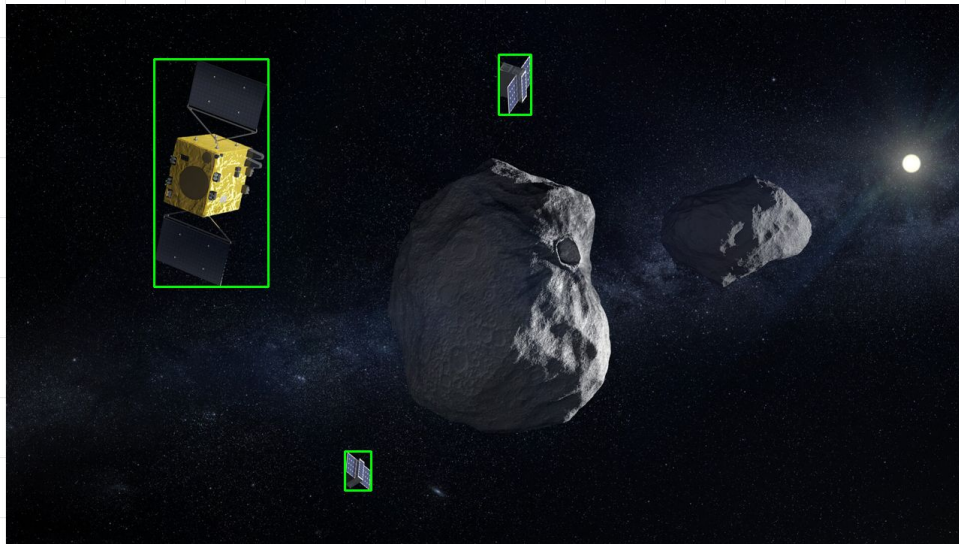
- Clasificación de Objetos
- Detección de Objetos
- Segmentación de Imágenes
- Reconocimiento facial
- OCR (Reconocimiento Óptico de Caracteres)
- Estimación de Poses
- Seguimiento de Objetos



Clasificación vs Detección de Objetos



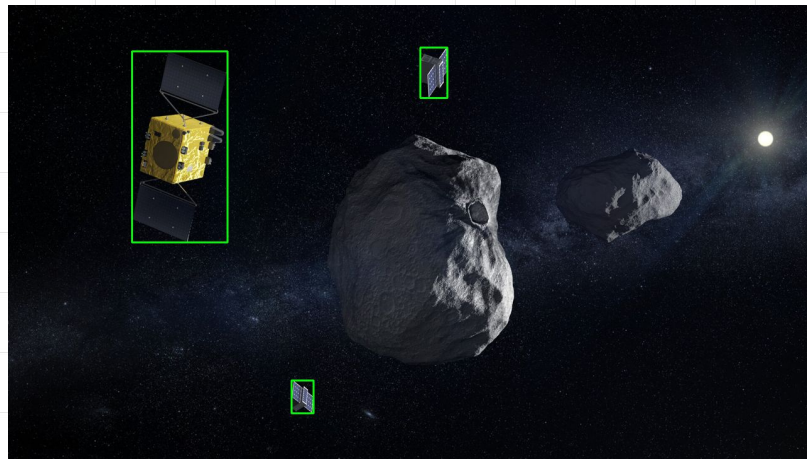
Clasificación vs Detección de Objetos



Clasificación vs Detección de Objetos

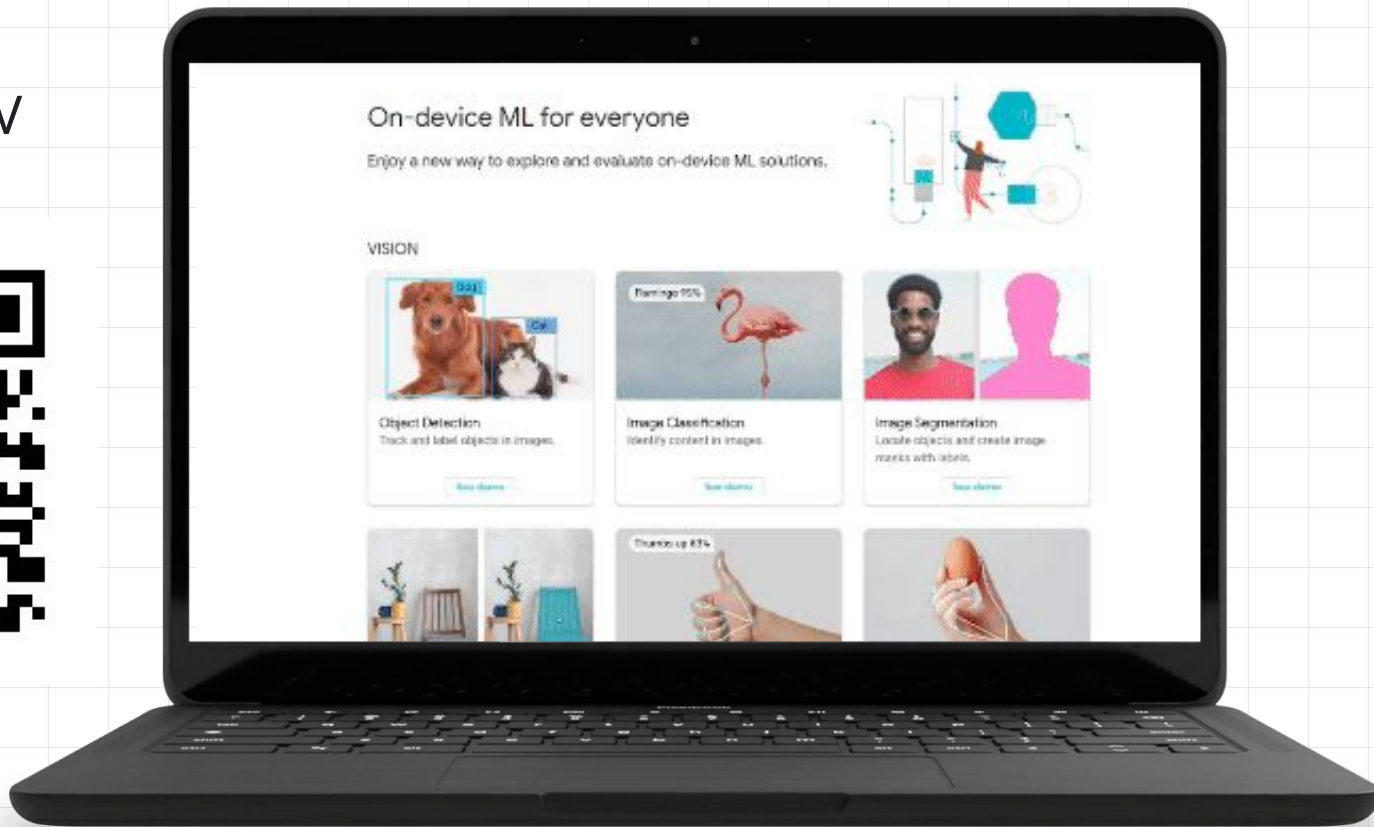


Clasificación

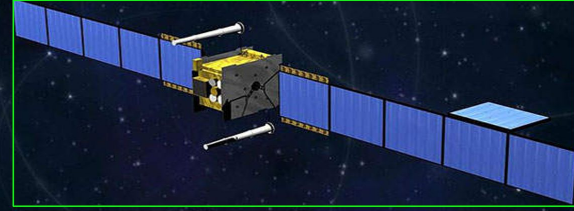
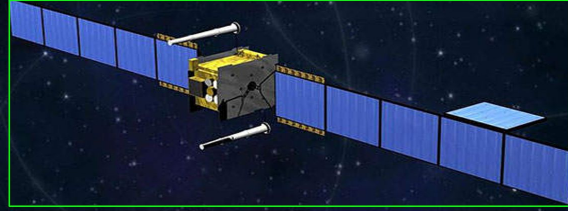
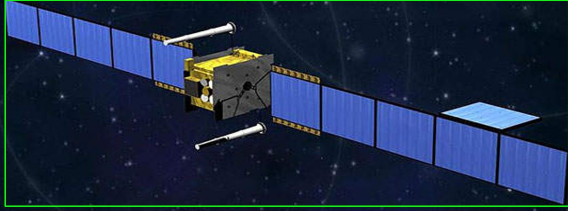


Detección

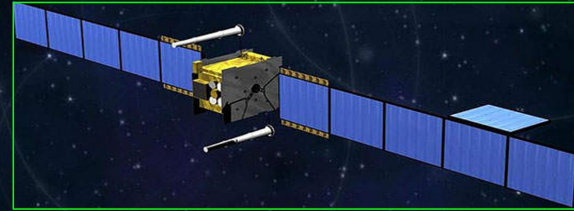
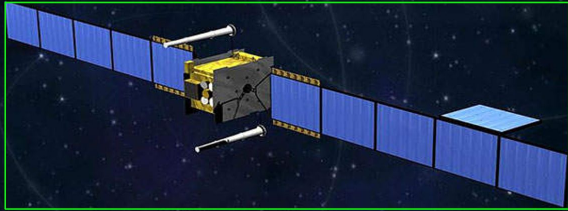
Tareas de CV MediaPipe



<https://mediapipe-studio.webapps.google.com/home>



Detección de objetos espaciales





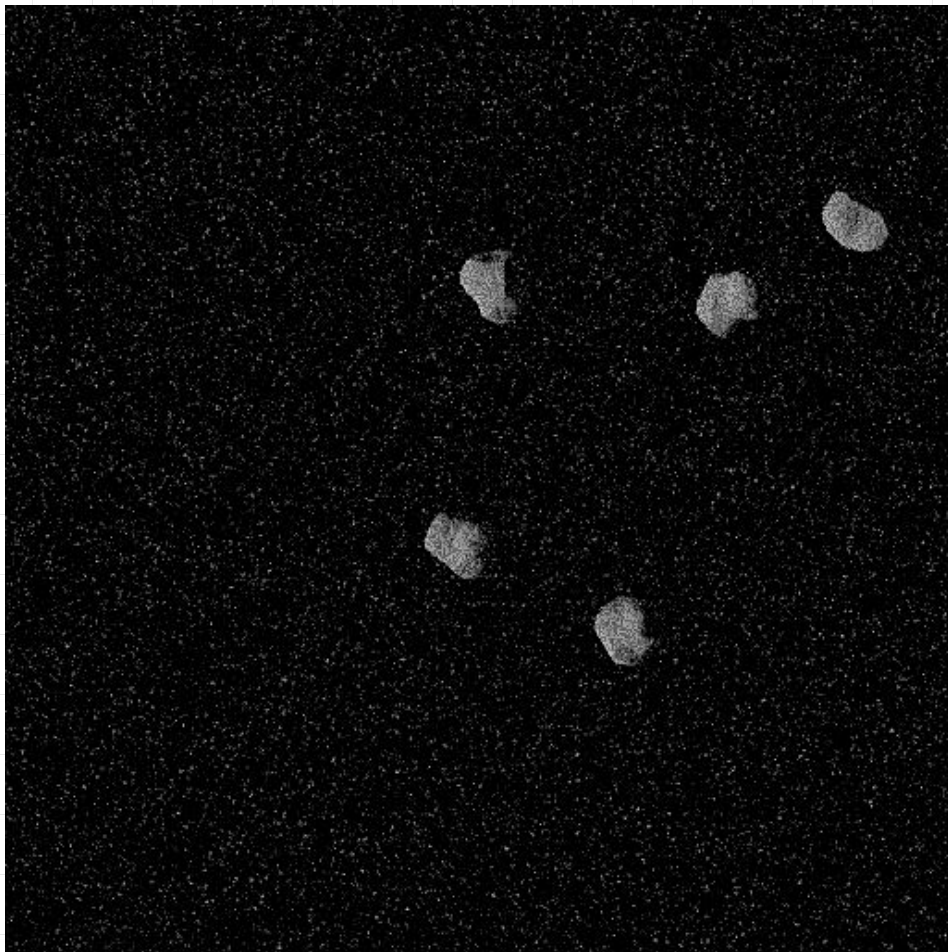
satellite-image-deep-learning

Datasets for deep learning applied to satellite and aerial imagery.

👉 satellite-image-deep-learning.com 👉

How to use this repository: if you know exactly what you are looking for (e.g. you have the paper name) you can `Control+F` to search for it in this page (or search in the raw markdown).

<https://github.com/satellite-image-deep-learning/datasets>



Files

Following files are available in the **resources** section:

- **train.zip** - (20000 samples) This zip file contains space with debris images, with images name corresponding to **ImageID** column of **train.csv**
- **train.csv** - (20000 samples) This csv file contains the **ImageID** column corresponding to **train.zip** and **bboxes** the column contains bounding boxes in a list.
- **val.zip** - (2000 samples) This zip file contains space with debris images, with images name corresponding to **ImageID** column of **val.csv**
- **val.csv** - (2000 samples) This csv file contains the **ImageID** column corresponding to **val.zip** and **bboxes** the column contains bounding boxes in a list.
- **test.zip** - (5000 samples) This zip file contains space with debris images which will be used to evaluate the performance of the model.

A Spacecraft Dataset for Detection, Segmentation and Parts Recognition

Hoang Anh Dung¹, Bo Chen², and Tat-Jun Chin²

¹The University of Adelaide

June 16, 2021

Abstract

Virtually all aspects of modern life depend on space technology. Thanks to the great advancement of computer vision in general and deep learning-based techniques in particular, over the decades, the world witnessed the growing use of deep learning in solving problems for space applications, such as self-driving robot, tracers, insect-like robot on cosmos and health monitoring of spacecraft. These are just some prominent examples that has advanced space industry with the help of deep learning. However, the success of deep learning models requires a lot of training data in order to have decent performance, while on the other hand, there are very limited amount of publicly available space datasets for the training of deep learning models. Currently, there is no public datasets for space-based object detection or instance segmentation, partly because manually annotating object segmentation masks is very time consuming as they require pixel-level labelling, not to mention the challenge of obtaining images from space. In this paper we aim to fill this gap by releasing a dataset for

ifications [6], just to name a few. Accompanying the development of space technologies is an increase in demand of space datasets, as most of state-of-art models for space technologies are using deep learning-based methods, which require a significant amount of annotated data for supervised training in order to have good performance. However, one challenge that hinders the advancement of these space technologies is the lack of publicly available datasets, due to sensitivity in the space area and the high cost of obtaining space-borne images.

One important technology in many space applications is the accurate localisation of space objects via visual sensor, such as object detection and segmentation in images, because localisation is a key step towards vision-based pose estimation which is critical for tasks like docking [7], servicing [8], or debris removal [9]. However, a severe challenge for space-based object detection and instance segmentation is the lack of accessible large datasets that have been well annotated. There has been some large scale segmentation dataset such as COCO[10], ImageNet [11], Pascal VOC [12] including masks of a large num-

¿Qué es una imagen?

Y (0)

X (1)



RGB

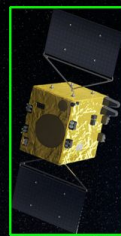
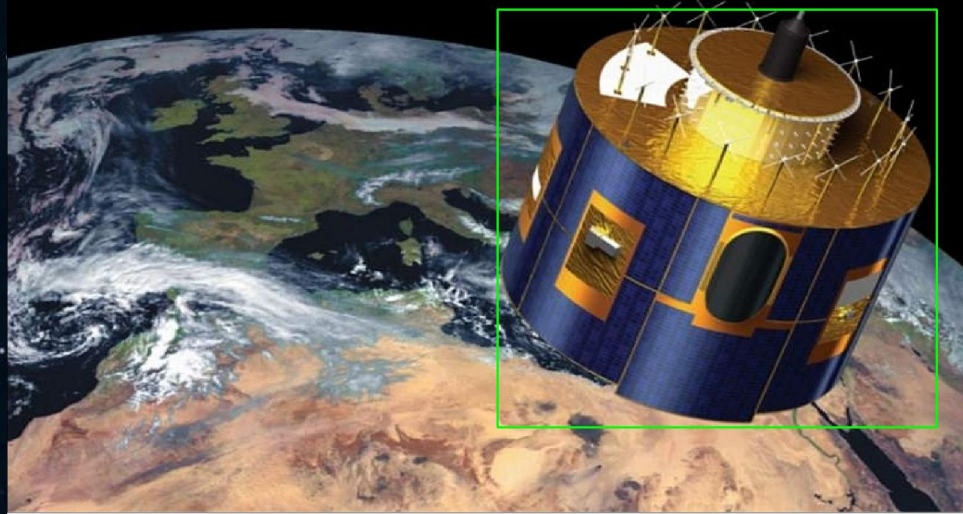
Posición	Clase	Valor
0	Alto (Height, número de filas)	720 píxeles
1	Ancho (Width, número de columnas)	1280 píxeles
2	Canales de color (Depth)	3 canales (R, G, B)

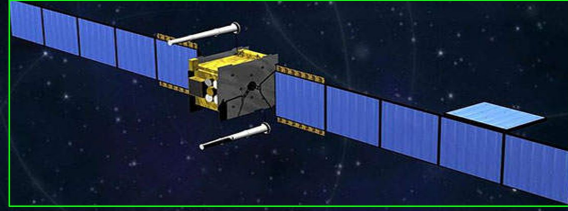
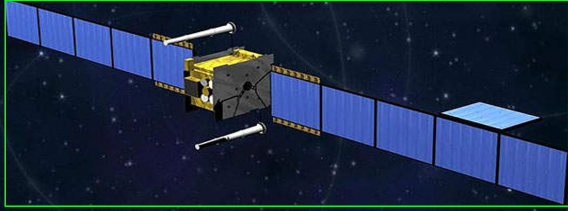
$X(1)$

$Y(0)$



$(0, 0) =$ 





Gracias
¿Preguntas?

